

On the Role the Friction Plays in an Ultra Precision Positioning System

Chen Hsieh and Kang-Chin Yang

Abstract—Friction, due to its destabilizing and drastic magnitude variation between stick and slip, is commonly considered as a detrimental factor to a positioning system. This paper, based on the new knowledge about pre-sliding dynamics, tries to put forward another point of view; it shows that the spring-damper-like property in the pre-sliding phase makes it possible to achieve the so called Precision Limit Position (PLP). With this kind of positioning, the position error and amount of in-position vibration of the system can remain to be equal to the resolution of the position sensor used in the feedback loop. This is, of course, the limit of the precision that a practical positioning system can achieve.

Index Terms—Precision-limit-positioning, Friction, Stribeck curve, nanotechnology, Aero-static bearings

I. INTRODUCTION

Researches on position control have become very popular over the past two decades. The standard for precision has also become more and more stringent. Just ten years ago, a motion table with sub-micro meter precision was considered as a high precision machine. But today people are talking about nanometer or even sub-nanometer precision. The resolution of the linear scale that is one of the most popular position measurement devices has also improved from micro-meter to sub-nanometer.

When it comes to precision positioning, people care about not only the steady state error and repeatability, they also ask about the performance of settling time, in-position vibration, step resolution, rigidity and many other indices according to the specific purpose for which the machine is implemented. In 2003, a stringent positioning standard called precision-limit-positioning (PLP) was proposed [1]. In addition to ultra high resolution, according to this standard, a zero-count steady state error with 100% repeatability is required for a digital positioning system. Furthermore, it also asks for a zero-count in-position vibration. Here the count is the minimum measurement unit of the digital position sensing device. In other words, PLP allows count-by-count operation. This is, of course, the limit the precision that a practical motion table can possibly achieve. To achieve this kind of performance, a thorough understanding of the system is

necessary and many factors have to be taken care of. Among these factors, how to eliminate the influence of friction might be one of the most important issues. Due to its highly nonlinear characteristics, time varying parameters and tremendous change in behavior between stick and slip, the friction is considered as one of the most unwanted things in a precision motion platform. Many researches have been done to compensate the friction effect. Due to the limited space they will not be repeated here. Some of them were reviewed in depth in [1]. These methods do improve the performance more or less. But few of them can fulfill the requirements of PLP. In fact, it is widely believed that an ultra precision motion table must use non-contact parts such as brushless motors and aero-static bearings. In this case the friction can almost be gotten rid of completely. Besides the property of frictionless, air bearings are also suitable for operation in a clean room. Furthermore, a well designed air bearing table can offer a very flat moving plane that can hardly be obtained using an ordinary linear guide or bearings that have contacting surfaces. This is an important property that is required by some extremely high end tables but is ignored in most applications. However, in spite of these advantages, it can be theoretically proved that the ultimate in-position stability that can be achieved is 1 count if there is no friction existing in the system. Zero-count in-position stability as required by PLP can never be accomplished for an air bearing system. In addition, this one-count stability, depending on the resolution, usually requires a driver with extremely high quality. Otherwise, significant residue vibration remains. It is noted that 1-count in-position vibration implies a triple vibration amount as compared with the zero-count case. Another drawback of the air bearing is that it has lower rigidity. To overcome these defects, some methods are available. The simplest way is to cut the air supply so that the moving table is stuck to the supporting plane. Or, a mechanical clamping device can be used to hold the sliding table when it gets to the desired position. But doing these ways reduce the system response speed if a repeated positioning process is required. Furthermore, these extra actions cause disturbances to the system and may ruin those benefits of an air bearing as mentioned earlier. In [2, 3] a special liquid material called Electrorheological fluid is used to generate the necessary damping as the slider gets close to the target. This is an innovative way, however, how to seal the liquid remains to be an open question.

C. Hsieh and Kang-Chin Yang are with the Department of Aeronautics and Astronautics, National Cheng Kung University, 1 University Rd. Tainan, Taiwan, Rep. of China. (E-mail: chsieh@mail.ncku.edu.tw). This work was supported in part by Grants 97-EC-17-A-05-S1-0014 and NSC 97-2221-E-006-078 from the Ministry of Economic Affairs and National Science Council, R.O.C.

It is obvious that the main drawback of an air bearing comes from that it lacks a damping mechanism as well as a restoring force. And these shortcomings cannot be completely made up using digital controls due to its quantization error. Ironically, the static friction that was treated as an annoying factor can offer quite good damping and restoring force if appropriately handled. As a matter of fact, this property of static friction has been applied in many positioning systems in the author's earlier and current research works to achieve the performance of PLP. In this paper, the effect of friction on position control will be further discussed from the dynamics point of view.

In the next section, the friction behavior will be briefly described. Some important properties will be summarized. Section 3 discusses the problems of most existing friction-model-based position controls. A simple but practical way to handle friction is then addressed. Two examples of ultra positioning systems are given in section 4 and followed by the conclusion section.

II. BEHAVIOR OF FRICTION

A. Infrastructure of Friction Mechanism

Friction might be one of the most difficult and oldest issues in the field of mechanics. This is mainly because that it is hard to observe the true contacting mechanism directly. Almost all conclusions about the mechanism of friction come from indirect observations or numerical simulations based on other theories. Among those descriptions about the contacting mechanism, the one given by Bowden and Tabor [4] and other researchers [5-8] might be most persuading. According to their description, during the pre-sliding motion, the contacting mechanism is, in fact, composed of junctions which are the regions where real contacts of minute asperities occur. Cold welding and strong adhesion take place at these points. This cold welding process was also indirectly proved by numerical simulations based on molecular dynamics and experiments using atomic force microscopes (AFM) [9, 10]. Bowden and Tabor made a statement about the source of friction. "Friction is the force required to shear intermetallic junctions plus the force required to plow the surface of the softer metal by asperities on the harder surface [4]." The friction force in the pre-sliding phase, on the surface, comes from the interaction of two totally different surfaces. It is, in fact, the reaction of one continuous material. This is why the pre-sliding behavior in an ordinary mechanism is so close to that of a plastic material. Hysteresis, creep, work hardening and even Bauchinger effect that consist of the behaviors of a plastic material can also be found in the behavior of pre-sliding motion. When those asperity junctions are broken, true slip occurs and the friction force is the so called kinetic friction and comes from the plowing motion of one plane on the other. It is obvious now, that the traditional Amontons-Coulomb model of friction which divides the

friction into exactly static and kinetic friction is too rough. Since even in the pre-sliding range, the sliding motion also exists. It is quite hard to distinguish which category the friction belongs to just by examining the relative velocity of these two surfaces. Besides the pre-sliding motion and slip motion, it is natural to speculate that there is still a transition motion where part of the junctions is broken and part of them is stuck. This happens when the relative motion is slow but persistent. It is noted that all junctions will break when the relative displacement is keeping increased but new junctions appear to adhere as long as the relative speed is slow enough to allow cold welding to occur. The observation for this kind of friction is even harder than those for the pre-sliding and slip phases, since it is very unstable in this phase. However, it is reasonable to assume that equilibrium between adhesion part and broken part will be achieved if a slow velocity can truly be maintained. The percent of adhesion part is going to decrease as the speed is increasing and the friction force is decreasing and finally becomes that of the kinetic friction. This kind of process is believed to yield the so called Stribeck effect.

B. Modeling of the Friction

A unified and rather complete mechanical model for friction has been constructed by the authors. This model is based on the details of friction mechanism as described above and has been proved to exhibit right friction behaviors under any given conditions. However, it is not appropriate for control design and is beyond the scope of this paper and, hence, will be described elsewhere. In this paper, the models for different phases are discussed separately.

B.1 Pre-sliding Phase

In [11], a model as shown in Fig. 1 is introduced. It is proved to possess all the pre-sliding behaviors described earlier. This model consists of three specially defined massless mechanical elements; they are the plastic module that has the properties of creep and work hardening, the nonlinear spring module that demonstrates the special Preisach hysteretic behavior and a standard damper. The governing equations for the plastic module and nonlinear spring module are described as follow.

Plastic module

Consider the plastic module in Fig. 1. Let σ be the force applied to the module and x_p be the deformation then we have

$$\begin{cases} \dot{x}_h = \begin{cases} \alpha(f(|\sigma|) - x_h), & \text{if } f(|\sigma|) > x_h \\ 0, & \text{otherwise} \end{cases} \\ \dot{x}_p = \text{sgn}(\sigma)\dot{x}_h \end{cases} \quad (1)$$

where α is a positive constant, $f(s)$ is a monotonically increasing function that satisfies $f(0) = 0$ and $\left. \frac{df}{ds} \right|_{s=0} = 0$.

In this model an extra state, x_h is introduced. It is a monotonically non-decreasing function standing for the amount of work hardening. Fig. 2 shows the typical static response of this model.

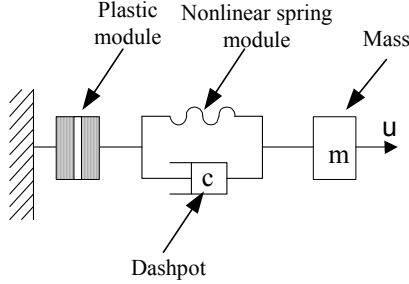


Fig. 1. The model for the pre-sliding phase

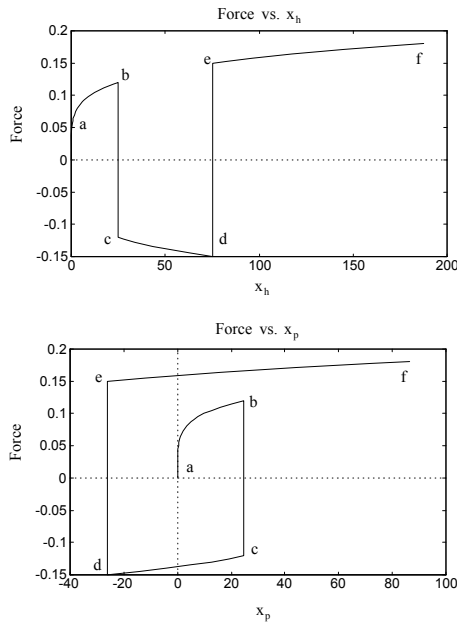


Fig. 2. Typical response of the plastic module

Nonlinear spring module

Consider the nonlinear spring module in Fig. 1. Let σ_s be the force applied to the spring and x_s be the elongation of the spring. Then the relation between them can be approximated by

$$\frac{d\sigma_s}{dx_s} = k_1 + k_2 e^{-\beta|x_s - x_r|} \quad (2)$$

where k_1 , k_2 and β are positive constants. x_r is the active reverse point of current curve branch. The reverse points follow a rule that carries out the memory and wipe-out properties of the hysteresis. Please see [11] for the detail. Fig. 3 is a typical response of the nonlinear spring.

The friction F equals the force σ or, equivalently, the summation of σ_s and the damping force $c\dot{x}_s$.

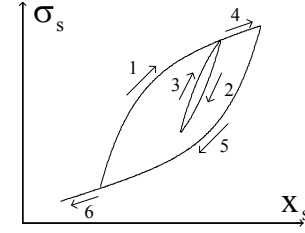


Fig. 3. Typical response of the nonlinear spring

B.2 Slip Phase

In the slip phase, the traditional model given below is appropriate

$$F = -\text{sgn}(v)(F_c + c_s(|v| - v_s)), \quad |v| > v_s \quad (3)$$

where F_c is the conventional kinetic friction and c_s is the damping coefficient. v_s is the maximum velocity where cold welding cannot completely develop to adhere any two contacting points.

B.3 Transition Phase

In this phase, the Stribeck effect dominates and can be approximated by

$$F = F_c + (F_s - F_c)e^{-\gamma(|v|/(v_s - |v|))^n}, \quad |v| \leq v_s, \quad (4)$$

where F_s is the maximum static friction. Roughly speaking, it is the force needed to break all conjunction points. γ and n are positive scalars.

As mentioned earlier that theoretically there is no clear criterion to determine which phase the friction is staying. It has been proven theoretically and experimentally that it is possible to achieve high speed while still stays in the pre-sliding phase. Many weird phenomena contradict to well known friction behaviors can be demonstrated. Therefore, it is not a correct way to judge the friction phase by just checking current velocity and displacement. However, in most positioning problems it is practical to say that it transfers from pre-sliding phase to transition phase if the friction force is greater than F_s . Conversely, it transfers back to the pre-sliding phase if the motion is slower than a specific speed. This speed can usually be “estimated” experimentally.

III. HOW TO HANDLE FRICTION IN A CONTROL SYSTEM

To compensate the friction force, the most common way is to estimate the friction based on a given model which is almost always in the form of a set of equations. However, the premise of this strategy is that a reasonable model structure is adopted. Unfortunately, this is not an easy task because of the following reasons. 1. When it comes to the precision, how to take care of the pre-sliding friction in the final stage of positioning is the crucial step. It has been proved in [11] that it is not possible to represent this phase of friction in the form of $\dot{x} = f(x, t, u)$ as is assumed in almost all nonlinear control text books. This is mainly because that the state of the

hysteresis in this phase should include all those reverse points in the hysteresis loop that have not been wiped out. Most existing friction model does not exhibit the hysteretic phenomenon and not mention to estimate the reverse point. Without knowing the correct reverse point, the friction force cannot be correctly estimated. 2. The friction parameters are time- and position-varying. Many investigators [12, 13] have found that it is very hard to get quantitatively repeatable results. This is true even for well-prepared metal-to-metal friction experiments [13], and not to mention the mechanisms in commercial machines. Experiment has shown that the maximum static friction may vary to the extent of 300% at the same point if the table moves away from it and back again. 3. Many friction parameters are hard to measure experimentally. For example, the Stribeck curve should be obtained from those experiments that conduct slow speed. Here the slow speed should be persistent slow speed rather than average slow speed. Due to its inherent highly destabilizing characteristic and the difficult to accurately measure extremely slow velocity in a digital and real-time way, it is quite difficult to conduct velocity control to achieve persistent and consistent slow speed. Most experiments, if not all, to our knowledge, can just accomplish average slow speed control. The curve obtained is, in fact, based on the average value of a stick-slip motion. This is far away from what it should be. 4. Some models still stick to old textbook concepts. The most well known concept is that the friction force is always opposite to the direction of motion. This kind of model is ok if the precision required is not high. However, it is detrimental for those systems asking for ultra high precision and in-position stability. In the pre-sliding phase, the friction force may be in the same direction of motion. Erroneously judging the friction direction implies compensation in wrong direction. Consequently, though many existing control strategies with friction compensation can indeed improve the precision, few of them can achieve zero-count in-position vibration as required by PLP.

Even though the friction has so many uncertain and nonlinear properties, it does have some important properties that can be utilized.

1. The static friction dynamics is temporarily time-invariant and locally position-invariant. By temporarily time-invariant we mean that the system properties remain constant during short interval of interest. This short interval is usually long enough for the positioning of each specific target point. By locally position-invariant we mean that the system properties remain unchanged if it stays in the pre-sliding range of any individual point.
2. The plastic module has a response much slower than that of the nonlinear spring. Though it still has some minor effects, it can usually be ignored in a practical case.
3. It turns out that, for control purposes, the main property of the nonlinear spring we concern is the lower bound of the slope of the hysteresis. Fortunately, this is a property that is

quite easy to measure. [11] has the detail on how to get these data. Furthermore, experience shows that this bound is quite consistent; it does not vary too much with time.

Based on these observations, it is clear that the closed loop system in the pre-sliding phase can, hence, be modeled as that shown in Fig. 4 if a PID control is adopted. It is noted that this is a time-invariant system with unknown but fixed system uncertainties. Furthermore, their bounds are known. It has been proved in [1] using Lyapunov theory for switched systems that the system is stable if

$$k_I < \frac{(c + k_D)(\underline{h} + k_p)}{m}$$

where $\underline{h}$ is the lower bound of the slope of the hysteresis branch of curves. It is interesting to note that this result is exactly the same for designing the PID control for a standard M-K-C system. As a matter of fact, experiments show that the behavior in the pre-sliding phase is very close to that of an M-K-C system. Since it is a type 0 system, to get zero steady state error, an integral type input should be included in the control loop. In fact, a single integral control in the pre-sliding phase can obtain a zero steady state error. This is different from the conclusion of some theoretical analyses where the Coulomb friction model is adopted. In that case, an integral control will cause a limit cycle. [14] As a matter of fact, if the target is in the pre-sliding range of the stick point, the friction plays the role of offering efficient proportional and derivative feedback in an analogue way. This is exactly what is lacking in a frictionless system and cannot be made up by a digital control loop. These results addressed here can be applied to direct drive systems as well as systems with reduction mechanisms.

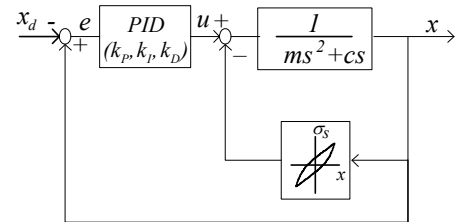


Fig. 4. Block diagram in the pre-sliding phase

IV. EXAMPLES OF ULTRA PRECISION POSITIONING

In this section, two systems are discussed. They are a rotational torque motor system and a linear table.

A. Position Control of a Torque Motor System

The system is shown in Fig. 5. The plant is a direct drive DC torque motor mounted on ordinary ball bearings. The motor is driven by a linear amplifier. Two optical encoders with different resolutions are mounted to measure the rotor position. The resolution of the coarse encoder is 86,400 count/rev or, equivalently, 15 arc-sec/count, while that of the fine encoder is 1,620,000 count/rev or, equivalently, 0.8 arc-sec/count. The rotor angle is measured simultaneously by

these two encoders. The outputs are compared to guarantee that the data obtained is correct. However, only one encoder is used in the feedback loop in each test.

Figures 6 and 7 show the dynamic responses of the system in the pre-sliding phase. We first pre-assign a desired rotor position and then gradually increase the current with a constant rate. The current is cut once the desired position is reached. Same procedure is repeated for 5 seconds. Typical response with a designated angle of 30 counts is shown in Fig. 6. It is clear that the behavior is very close to that of a standard M-K-C system. In Fig. 7, instead of cutting the current, the final current is held once the designated angle is reached. Here the pre-assigned angle is 15 counts. The creep and work hardening phenomenon is obvious. A typical torque-versus-displacement curve is shown in Fig. 8. It shows a special Preisach hysteresis, in which the congruency property is good for both ways. [15] These properties can be faithfully simulated by the model shown in Fig. 1. To design the position control system, the lower bound of the hysteresis loop is measured. To do this, we follow the process described in [11] to do static response tests every 30 degrees. This procedure is repeated for 3 circles. The nonlinear spring response is then extracted from the data and is shown in Fig. 9. The lower bound $\underline{h}$ is estimated to be $\underline{h}=110$ Nm/rad. It is clear that this stands for a stiff spring. Since the dynamics of the pre-sliding phase is completely different from that of the slip phase. To obtain good response, different control strategies are designed for different phases. Fig. 10 shows some typical step responses. To verify the repeatability, more than 4 million consecutive positioning tests using the fine encoder have been conducted. In all these tests, only one positioning failed and had an error of 2 counts left.

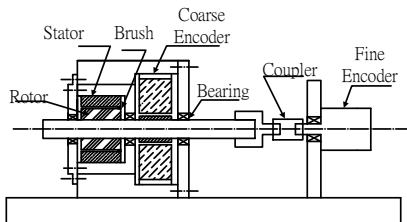


Fig. 5. Schematic diagram of the torque motor system

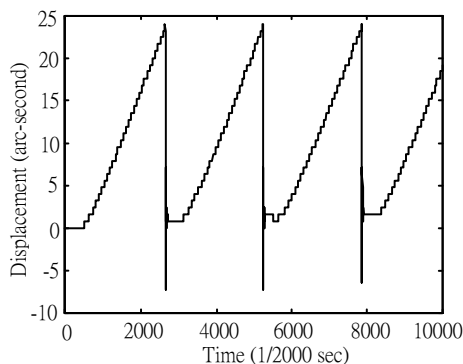


Fig. 6. Dynamic response in the pre-sliding phase (Test 1)

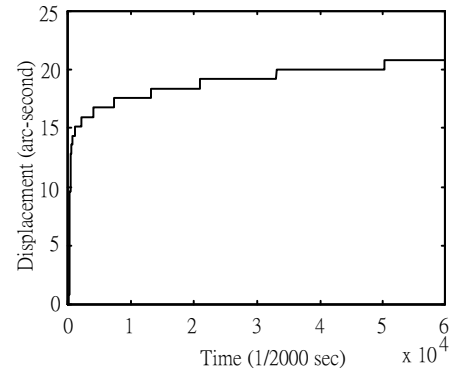


Fig. 7. Dynamic response in the pre-sliding phase (Test 2)

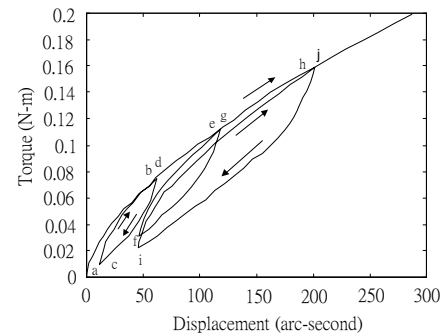


Fig. 8. The static response in the pre-sliding phase

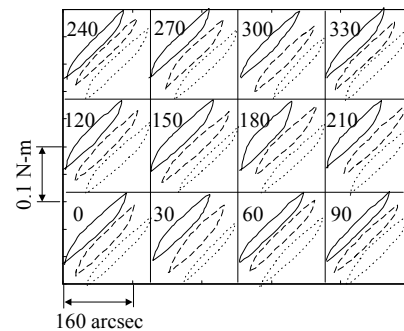


Fig. 9. Response of the nonlinear spring at different positions

B. Position Control of a Linear Motor

The system used in this case is an ordinary two-axis linear motor table with one axis stacking on the other. The motors are manufactured by HIWIN Mikrosystem Corp. The traveling distance of each axis is 270 mm. On each axis, it is mounted a Heidenhain linear scale (LIP481), which has two harmonic analogue outputs with a resolution of $2\mu\text{m}/\text{pitch}$ for each phase. Using electronic interpolation we can magnify the resolution by 1000 times and get a position measurement with a resolution of 2 nm/count. Due to the limit of the electronic circuit, the maximum speed allowed in this case is 30 mm/sec. The bearings used in this system are ordinary linear guides with friction in it. In this paper, only the upper axis is discussed. A typical positioning with a distance of 200mm is shown in Fig. 11. Since the table is put directly on the ground without any shock absorbing device under it.

Small in-position vibration due to environment disturbances exists. But the vibration is limited to 2 counts at most. Fig. 12 shows a count-by-count operation. In this test, the table moves 1 count every time the operator hits the arrow keys.

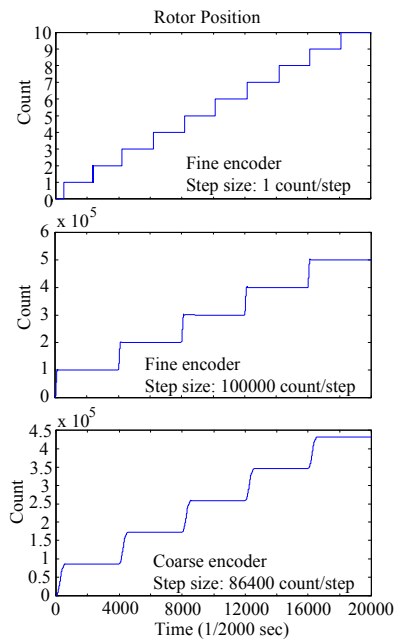


Fig. 10. Step responses of the rotor system

V. CONCLUSION

For a positioning system with sub-micro, nano or even pico meter precision, it is usually subject to significant in-position vibration. This situation cannot be eliminated without extra locking devices if the parts used in the system are frictionless. However, it can be reduced or even totally eliminated for a system with friction if the resolution of the control effort is fine enough. No further locking devices are required. This result is quite different from what people used to believe. Of course, frictionless bearings such as air bearings do have some superior properties that a ball bearing does not possess. But achieving high precision positioning is definitely not the reason we want to use such kind of bearings.

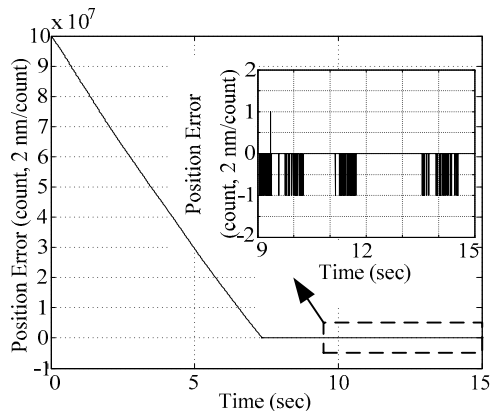


Fig. 11. A typical response of 200mm positioning

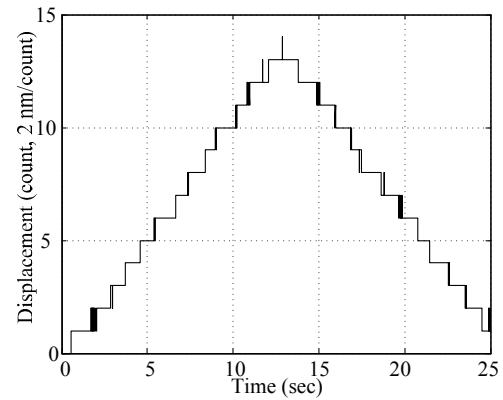


Fig.12. Typical count-by-count operation

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